Week 6: Linear spaces and subspaces

Linear Algebra I

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These notes were drafted with the help of ChatGPT from my handwritten notes and the discussion used to prepare the recitation.

1. What makes a set a linear space?

So far, we have worked mainly with vectors and matrices. Their entries let us add objects and multiply them by scalars, and many of our arguments use only these two operations. We now look at the same structure in other settings. Polynomials, functions, and infinite sequences will all serve as vectors. Throughout these notes, the scalars are real numbers.

A *real linear space*, also called a *real vector space*, is a set V with addition and scalar multiplication satisfying the usual rules. Addition is associative and commutative, there is a zero vector 0, and every vector has an additive inverse. For $u, v \in V$ and $a, b \in \mathbb{R}$, scalar multiplication satisfies

$$a(u+v) = au + av, \quad (a+b)u = au + bu,$$
$$a(bu) = (ab)u, \qquad\qquad 1u = u.$$

The word "vector" refers to an element of such a space. It need not be a column of numbers. For example, the zero vector in a function space is the function that is zero at every point.

The subspace test. A subset $W \subseteq V$ is a *subspace* if it is a linear space with the operations inherited from V. To check this, it is enough to show that $W \neq \varnothing$ and

$$au + bv \in W \qquad \text{for every } u, v \in W \text{ and } a, b \in \mathbb{R}.$$

Indeed, this condition gives closure under addition and scalar multiplication. Taking both scalars to be zero shows that $0 \in W$, and taking a scalar to be -1 gives additive inverses. All the other rules are inherited from V.

Equivalently, one may check three things separately: $0 \in W$, $u + v \in W$ whenever $u, v \in W$, and $au \in W$ whenever $u \in W$ and $a \in \mathbb{R}$. Merely containing zero is not enough, but it is often the quickest first check.

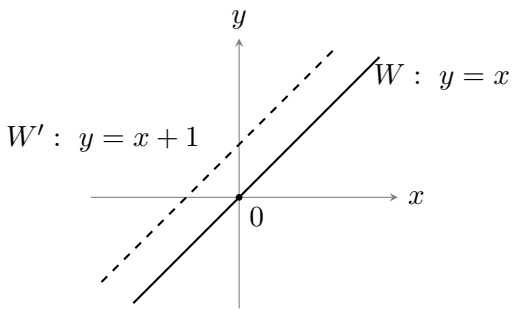
A line and its translate. In $\mathbb{R}^2$, the line

$$W = \{(t, t) : t \in \mathbb{R}\}$$

is a subspace because $a(s, s) + b(t, t) = (as + bt, as + bt)$. The parallel line

$$W' = \{(t, t+1) : t \in \mathbb{R}\}$$

is not a subspace: it does not contain $(0, 0)$. Both sets are straight lines, but only the first passes through the origin.



2. Subspaces of matrix spaces

Let $M_{m,n}(\mathbb{R})$ be the set of real $m \times n$ matrices. Entrywise addition and scalar multiplication make it a linear space. Its zero vector is the zero matrix. Once the matrix size is fixed, we can ask which additional conditions on the entries define subspaces.

Upper and lower triangular matrices. For square matrices, set

$$U_n = \{A = (a_{ij}) \in M_n(\mathbb{R}) : a_{ij} = 0 \text{ when } i > j\},$$
$$L_n = \{A = (a_{ij}) \in M_n(\mathbb{R}) : a_{ij} = 0 \text{ when } i < j\}.$$

For example, their elements for $n = 3$ have the respective forms

$$\begin{pmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{pmatrix}, \qquad \begin{pmatrix} a & 0 & 0 \\ b & c & 0 \\ d & e & f \end{pmatrix}.$$

If $A, B \in U_n$, then every entry below the diagonal of $sA + tB$ is $s \cdot 0 + t \cdot 0 = 0$. Thus U_n is a subspace. The same argument applies to L_n. The diagonal matrices form another subspace, consisting of the matrices that belong to both U_n and L_n.

Symmetric and skew-symmetric matrices. Recall that A^{T} denotes the transpose. We define

$$\mathcal{S}_n = \{A \in M_n(\mathbb{R}) : A^{\mathsf{T}} = A\},$$
$$\mathcal{K}_n = \{A \in M_n(\mathbb{R}) : A^{\mathsf{T}} = -A\}.$$

Both contain the zero matrix. If $A, B \in \mathcal{S}_n$, then

$$(sA + tB)^{\mathsf{T}} = sA^{\mathsf{T}} + tB^{\mathsf{T}} = sA + tB.$$

If $A, B \in \mathcal{K}_n$, the same calculation gives

$$(sA + tB)^{\mathsf{T}} = -sA - tB = -(sA + tB).$$

Hence both sets are subspaces. In the 2×2 case, their elements have the forms

$$\begin{pmatrix} a & b \\ b & c \end{pmatrix} \qquad \text{and} \qquad \begin{pmatrix} 0 & d \\ -d & 0 \end{pmatrix}.$$

In particular, a real skew-symmetric matrix has zero diagonal: the relation $a_{ii} = -a_{ii}$ forces $a_{ii} = 0$.

These examples are defined by homogeneous linear conditions on the entries. Conditions of a different kind need not give subspaces. For instance, the invertible matrices do not form a subspace of $M_n(\mathbb{R})$, since the zero matrix is not invertible. Later we will see that every matrix splits into a symmetric part and a skew-symmetric part.

3. Polynomial spaces

A polynomial can also be added to another polynomial and multiplied by a real number. Let

$$\mathbb{R}[x] = \{a_0 + a_1 x + \cdots + a_m x^m : m \geq 0,\ a_i \in \mathbb{R}\}.$$

Every individual polynomial has only finitely many terms, although there is no single degree bound for all of $\mathbb{R}[x]$. With the usual operations, $\mathbb{R}[x]$ is a linear space.

For a fixed integer $n \geq 0$, we define

$$P_n = \{a_0 + a_1 x + \cdots + a_n x^n : a_0, \ldots, a_n \in \mathbb{R}\}.$$

This is the space of polynomials of degree at most n, together with the zero polynomial. Writing the definition in terms of coefficients avoids any issue about the degree of zero.

If $p, q \in P_n$, then $sp + tq$ still has no term of degree greater than n. Thus P_n is a subspace of $\mathbb{R}[x]$. For example,

$$P_2 = \{a + bx + cx^2 : a, b, c \in \mathbb{R}\}.$$

The correspondence $a + bx + cx^2 \leftrightarrow (a, b, c)$ respects addition and scalar multiplication. In this sense, working with P_2 is much like working with $\mathbb{R}^3$.

Why "degree exactly n" is different. The set

$$S_n = \{p \in \mathbb{R}[x] : \deg p = n\}$$

is not a subspace. It does not contain the zero polynomial. It also fails closure under addition: x^n and $-x^n$ belong to S_n, but their sum is zero. For $n \geq 1$, cancellation can even produce a nonzero polynomial of lower degree, as in

$$(x^n + 1) + (-x^n) = 1.$$

The condition "degree at most n" allows cancellation of leading terms, whereas "degree exactly n" does not.

4. Function spaces and pointwise operations

The polynomial examples suggest a larger setting. Let $\mathcal{F}(\mathbb{R}, \mathbb{R})$ be the set of all functions from $\mathbb{R}$ to $\mathbb{R}$. Define addition and scalar multiplication *pointwise*:

$$(f + g)(x) = f(x) + g(x), \qquad (af)(x) = af(x).$$

The vector space rules follow from the corresponding rules for real numbers at each point x. The zero vector is the zero function, and the additive inverse of f is the function $-f$.

Regularity conditions. Several familiar classes of functions are subspaces:

$$\begin{aligned} C(\mathbb{R}) &= \{f : \mathbb{R} \to \mathbb{R} : f \text{ is continuous}\}, \\ C^k(\mathbb{R}) &= \{f : \mathbb{R} \to \mathbb{R} : f \text{ has } k \text{ continuous derivatives}\}, \\ C^\infty(\mathbb{R}) &= \{f : \mathbb{R} \to \mathbb{R} : f \text{ has continuous derivatives of all orders}\}. \end{aligned}$$

Here $k \geq 1$. The notation C^k requires continuity of the derivatives through order k, not just their existence. Smooth functions are the elements of C^∞.

A linear combination of continuous functions is continuous. Similarly, if $f, g \in C^k(\mathbb{R})$, then, for $0 \leq j \leq k$,

$$(af + bg)^{(j)} = af^{(j)} + bg^{(j)},$$

and the right-hand side is continuous. This proves closure for $C^k(\mathbb{R})$, and the same argument works for $C^\infty(\mathbb{R})$. All three spaces contain the zero function.

Even and odd functions. Within $\mathcal{F}(\mathbb{R}, \mathbb{R})$, let

$$\begin{aligned} E &= \{f : f(-x) = f(x) \text{ for every } x \in \mathbb{R}\}, \\ O &= \{f : f(-x) = -f(x) \text{ for every } x \in \mathbb{R}\}. \end{aligned}$$

For even functions f, g, one has

$$(af + bg)(-x) = af(x) + bg(x) = (af + bg)(x).$$

For odd functions, the same calculation gives the negative of $(af + bg)(x)$. Since the zero function is both even and odd, E and O are subspaces. Examples include $1, x^2, \cos x$ in E and $x, x^3, \sin x$ in O.

Functions with a fixed period. Fix a number $T > 0$. The set

$$\mathcal{P}_T = \{f \in \mathcal{F}(\mathbb{R}, \mathbb{R}) : f(x+T) = f(x) \text{ for every } x \in \mathbb{R}\}$$

is a subspace, since

$$(af + bg)(x+T) = af(x+T) + bg(x+T) = (af + bg)(x).$$

Here T is a period, and need not be the smallest positive period. Constant functions, including zero, belong to $\mathcal{P}_T$.

Fixing a common period matters. The set of all periodic functions, with the period allowed to depend on the function, is not closed under addition. For example, $\sin x$ and $\sin(\sqrt{2}\,x)$ are periodic, but their sum is not. To see this, suppose $h(x) = \sin x + \sin(\sqrt{2}\,x)$ had a period $T > 0$. Its derivatives would have the same period, so

$$h'' + h = -\sin(\sqrt{2}\,x)$$

would also be T-periodic. It follows that $\sin x$ would be T-periodic as well. A positive period of $\sin(\omega x)$, with $\omega > 0$, must satisfy $\omega T \in 2\pi\mathbb{Z}$. Thus both T and $\sqrt{2}\,T$ would be integer multiples of 2π, contradicting the irrationality of $\sqrt{2}$.

5. Integral constraints and differential equations

The examples above use pointwise conditions or regularity. We can also define subspaces using an integral or a differential equation. The essential feature is again linearity.

A zero-integral condition. In $C([0,1])$, consider

$$W = \left\{ f \in C([0,1]) : \int_0^1 f(x)\,dx = 0 \right\}.$$

The zero function belongs to W. If $f, g \in W$ and $a, b \in \mathbb{R}$, linearity of the integral gives

$$\int_0^1 (af + bg)(x)\,dx = a\int_0^1 f(x)\,dx + b\int_0^1 g(x)\,dx = 0.$$

Hence W is a subspace. For instance, $f(x) = x - \frac{1}{2}$ belongs to W.

If the right-hand side of the condition is changed to 1, the resulting set is not a subspace, because it does not contain zero. More precisely,

$$\left\{ f \in C([0,1]) : \int_0^1 f(x)\,dx = 1 \right\} = 1 + W.$$

Here $1 + W$ denotes the set of functions $1 + g$ with $g \in W$, and 1 is the constant function. This is the function-space version of a line translated away from the origin.

A homogeneous differential equation. Fix real coefficients a, b, c, with $a \neq 0$, and consider the solution set in $C^2(\mathbb{R})$ of

$$ay'' + by' + cy = 0.$$

This is a homogeneous linear equation: every term depends linearly on the unknown function. If y_1, y_2 are solutions, then

$$\begin{aligned} a(sy_1 + ty_2)'' + b(sy_1 + ty_2)' + c(sy_1 + ty_2) &= s(ay_1'' + by_1' + cy_1) \\ &\quad + t(ay_2'' + by_2' + cy_2) = 0. \end{aligned}$$

Together with the zero solution, this proves that the solution set is a subspace.

For the equation $y'' + y = 0$, every solution has the form

$$y(x) = A\cos x + B\sin x, \qquad A, B \in \mathbb{R}.$$

Direct differentiation verifies that these functions are solutions. To see that there are no others, start with any solution y and set

$$z(x) = y(x) - y(0)\cos x - y'(0)\sin x.$$

Then $z'' + z = 0$ and $z(0) = z'(0) = 0$. Consequently,

$$\frac{d}{dx}(z(x)^2 + z'(x)^2) = 2z'(x)(z(x) + z''(x)) = 0.$$

The expression in parentheses is constant and is zero at $x = 0$. Thus $z = 0$, proving the claimed description of the solution set.

A nonhomogeneous differential equation. The constant function $y = 1$ is a particular solution of $y'' + y = 1$. Subtracting it from any other solution gives a solution of the homogeneous equation. Therefore all solutions are

$$y(x) = 1 + A\cos x + B\sin x, \qquad A, B \in \mathbb{R}.$$

This set is not a subspace, since the zero function does not solve the equation. It is a translate of the homogeneous solution space, just as in the integral example.

6. Spaces of infinite sequences

An infinite sequence is another kind of vector. We use $\mathbb{N} = \{1, 2, 3, \ldots\}$ and write

$$\mathbb{R}^{\mathbb{N}} = \{(a_1, a_2, a_3, \ldots) : a_n \in \mathbb{R}\}.$$

Addition and scalar multiplication are coordinatewise:

$$(a_n) + (b_n) = (a_n + b_n), \qquad \lambda(a_n) = (\lambda a_n).$$

These operations make $\mathbb{R}^{\mathbb{N}}$ a linear space, whose zero vector is $(0, 0, 0, \ldots)$.

Bounded sequences. Let

$$\ell^\infty = \{(a_n) \in \mathbb{R}^{\mathbb{N}} : \sup_{n\geq 1} |a_n| < \infty\}.$$

If $|a_n| \leq M$ and $|b_n| \leq N$ for every n, then

$$|sa_n + tb_n| \leq |s|M + |t|N \qquad \text{for every } n.$$

Thus every linear combination of two bounded sequences is bounded. Since the zero sequence is bounded, ℓ^∞ is a subspace of $\mathbb{R}^{\mathbb{N}}$. The bound may depend on the sequence; no common bound is imposed on all elements of the space.

By contrast, the set $\{(a_n) : |a_n| \leq 1 \text{ for all } n\}$ is not a subspace: it contains the constant sequence $(1, 1, \ldots)$, but not twice that sequence.

Convergent sequences. Let c be the set of real sequences with a finite real limit. If $a_n \to A$ and $b_n \to B$, then

$$sa_n + tb_n \longrightarrow sA + tB.$$

This proves that c is a subspace. The limit is allowed to vary from one sequence to another.

The sequences converging to zero form a smaller subspace,

$$c_0 = \{(a_n) \in \mathbb{R}^{\mathbb{N}} : a_n \to 0\}.$$

By the same limit rule, c_0 is closed under linear combinations. In contrast, the sequences converging to 1 do not form a subspace: the zero sequence is missing, and the sum of two such sequences converges to 2.

Every convergent sequence is bounded, so

$$c_0 \subsetneq c \subsetneq \ell^\infty \subsetneq \mathbb{R}^{\mathbb{N}}.$$

The constant sequence $(1, 1, \ldots)$ belongs to c but not c_0; the sequence $((-1)^n)$ is bounded but not convergent; and (n) is unbounded. These examples show why the inclusions are strict. The sequence $(1/n)$ is a nonzero element of c_0.

7. Kernels and images of linear maps

We can now describe a common source of subspaces. Let V and Z be real linear spaces. A map $T : V \to Z$ is *linear* if

$$T(au + bv) = aT(u) + bT(v) \qquad (u, v \in V,\ a, b \in \mathbb{R}).$$

Taking both scalars to be zero gives $T(0) = 0$. The *kernel* and *image* of T are

$$\ker T = \{v \in V : T(v) = 0\}, \qquad \operatorname{im} T = \{T(v) : v \in V\}.$$

The kernel is a subset of the domain V, while the image is a subset of the target Z.

Why both are subspaces. If $u, v \in \ker T$, then

$$T(au + bv) = aT(u) + bT(v) = 0,$$

so $au + bv \in \ker T$. Also $0 \in \ker T$. For the image, if $z_1 = T(u)$ and $z_2 = T(v)$, then

$$az_1 + bz_2 = T(au + bv) \in \operatorname{im} T.$$

Since $0 = T(0)$ is in the image, $\operatorname{im} T$ is also a subspace.

A matrix example. For $A \in M_{m,n}(\mathbb{R})$, the map $T_A : \mathbb{R}^n \to \mathbb{R}^m$, defined by $T_A(x) = Ax$, is linear. Its kernel is the solution space of $Ax = 0$, and its image is the column space of A. For example, if

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad T_A(x_1, x_2, x_3) = (x_1 + x_2, x_3),$$

then

$$\ker T_A = \{(t, -t, 0) : t \in \mathbb{R}\}, \qquad \operatorname{im} T_A = \mathbb{R}^2.$$

The first equality follows by solving $x_1 + x_2 = 0$ and $x_3 = 0$. For the second, any $(u, v) \in \mathbb{R}^2$ equals $T_A(u, 0, v)$.

Earlier examples as kernels. The integration map and the differential operator

$$\begin{aligned} I : C([0,1]) &\longrightarrow \mathbb{R}, \qquad I(f) = \int_0^1 f(x)\,dx, \\ L : C^2(\mathbb{R}) &\longrightarrow C(\mathbb{R}), \quad L(y) = y'' + y \end{aligned}$$

are linear. Their kernels are precisely the zero-integral functions and the solutions of $y'' + y = 0$. Likewise, the limit map $c \to \mathbb{R}$, $(a_n) \mapsto \lim a_n$, is linear and has kernel c_0.

There is also a general explanation for the translated solution sets. If $T(v_0) = z$, then

$$\{v \in V : T(v) = z\} = v_0 + \ker T.$$

Indeed, $T(v) = z$ is equivalent to $T(v - v_0) = 0$. Thus any nonempty solution set of a linear equation is a translate of its homogeneous solution space. When $z \neq 0$, it cannot be a subspace because it does not contain zero.

8. Intersections, sums, and unions

Let U and W be subspaces of the same linear space V. We can form new sets from them, but the operation matters.

The intersection. The intersection $U \cap W$ is a subspace. It contains zero. If $u, v \in U \cap W$, then $au + bv$ lies in U because U is a subspace, and it lies in W for the same reason. Hence it lies in their intersection.

For example, a matrix is both upper and lower triangular exactly when it is diagonal. Therefore $U_n \cap L_n$ is the space of diagonal matrices.

The sum. The sum of the two subspaces is defined by

$$U + W = \{u + w : u \in U,\ w \in W\}.$$

This set contains zero. If $x_1 = u_1 + w_1$ and $x_2 = u_2 + w_2$, then

$$ax_1 + bx_2 = (au_1 + bu_2) + (aw_1 + bw_2) \in U + W.$$

Thus $U + W$ is a subspace. It contains both U and W, since one may take the other summand to be zero. Moreover, any subspace containing both must contain every $u + w$. In this precise sense, $U + W$ is the smallest subspace containing U and W.

A calculation in $\mathbb{R}^3$. Consider

$$U = \{(a, b, 0) : a, b \in \mathbb{R}\}, \qquad W = \{(0, c, d) : c, d \in \mathbb{R}\}.$$

These are the xy-plane and the yz-plane. A vector belongs to both precisely when its first and third coordinates vanish, so

$$U \cap W = \{(0, t, 0) : t \in \mathbb{R}\}.$$

On the other hand, every $(x, y, z) \in \mathbb{R}^3$ can be written as

$$(x, y, z) = (x, y, 0) + (0, 0, z),$$

with the first summand in U and the second in W. Thus $U + W = \mathbb{R}^3$. Such a decomposition is not unique: for any $t \in \mathbb{R}$, we also have

$$(x, y, z) = (x, y - t, 0) + (0, t, z).$$

The freedom comes from their common vectors on the y-axis.

Why the union is different. The union $U \cup W$ consists of vectors belonging to at least one of the two spaces. It need not contain their sums. For example, take the two coordinate axes in $\mathbb{R}^2$. Their union contains $(1, 0)$ and $(0, 1)$, but not $(1, 1)$. Their sum, however, is all of $\mathbb{R}^2$.

In fact, $U \cup W$ is a subspace exactly when $U \subseteq W$ or $W \subseteq U$. One direction is immediate. For the other, suppose neither inclusion holds, and choose $u \in U \setminus W$ and $w \in W \setminus U$. If $u + w \in U$, then $w = (u + w) - u \in U$, a contradiction. If $u + w \in W$, the same argument gives $u \in W$. Thus $u + w \notin U \cup W$, so the union is not closed under addition.

9. Two useful decompositions

The sum construction lets us return to the matrix and function examples. In both cases, every object can be written as a sum of two special types, and the expression is unique.

First observe a general fact. If $U \cap W = \{0\}$, then a representation $v = u + w$, with $u \in U$ and $w \in W$, is unique whenever it exists. Indeed, if also $v = u' + w'$, then

$$u - u' = w' - w \in U \cap W.$$

Hence both differences are zero. When $V = U + W$ and $U \cap W = \{0\}$, we call this a *direct sum* and write $V = U \oplus W$.

Symmetric and skew-symmetric parts. For any $A \in M_n(\mathbb{R})$, define

$$S = \frac{A + A^{\mathsf{T}}}{2}, \qquad K = \frac{A - A^{\mathsf{T}}}{2}.$$

Then $S^{\mathsf{T}} = S$, $K^{\mathsf{T}} = -K$, and $A = S + K$. A matrix that is both symmetric and skew-symmetric satisfies $A = -A$, hence $A = 0$. Therefore

$$M_n(\mathbb{R}) = \mathcal{S}_n \oplus \mathcal{K}_n.$$

For instance,

$$\begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 3 & 3 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

The two matrices on the right are the unique symmetric and skew-symmetric parts.

Even and odd parts. For any function $f : \mathbb{R} \to \mathbb{R}$, define

$$f_e(x) = \frac{f(x) + f(-x)}{2}, \qquad f_o(x) = \frac{f(x) - f(-x)}{2}.$$

Substituting $-x$ shows that f_e is even and f_o is odd, and their sum is f. A function that is both even and odd satisfies $f(x) = -f(x)$ at every point, so it is zero. Thus

$$\mathcal{F}(\mathbb{R}, \mathbb{R}) = E \oplus O.$$

For $f(x) = 1 + x + x^2$, the decomposition is $f_e(x) = 1 + x^2$ and $f_o(x) = x$. If f is continuous or belongs to $C^k(\mathbb{R})$, both parts have the same regularity, so the decomposition also works within those spaces.

These formulas come from the same simple operation: average an object with its transpose or reflection, and then take half their difference. The subspace language explains both why the two parts stay in their respective spaces and why the decomposition is unique.